

Executive Summary (See ~results/visuals/logistic_regression_analysis.png, etc.)

Introduction:

This executive summary presents an analysis of customer churn from a subscription-based service. These insights come from applying advanced machine learning models to understand the primary drivers in customer churning. By leveraging Logistic Regression, Random Forest, and XGBoost classifiers, this project identifies the features that most strongly predict churn.

Churn is Most Affected By:

- **Tenure:** Shorter tenure strongly increases the likelihood of churn across all models
- **Contract Type:** Customers with one or two year contracts are much more likely to churn than those on month-to-month contracts
- **Internet Service - Fiber Optic:** Customers with fiber optic service are at higher risk of churn
- **Online Security & Tech Support:** Absence of these significantly elevates risk of churn

Methods to Reduce Churn:

Targeted Retention Initiatives: Focus retention campaigns on customers with short tenure and on month-to-month contracts. Encourage migration to longer-term contracts with loyalty points or reward bundles.

Fiber Optic Upsell: Customers with fiber optic service may have higher service expectations. Improve service reliability and offer tailored upsell packages.

Value-Add Services: Offer discounted services or free periods for online security and tech support to customers without these features. This might increase customer loyalty.

Conclusion

This analysis highlights tenure, contract type, and service attributes as key factors to identifying churn risk. By strategically targeting high-risk customers and optimizing service offerings based on these insights, the company will be well positioned to reduce churn, improve customer satisfaction, and strengthen competitive edge in the marketplace.